

Wearable Technology Gaining Ground, Thanks to Smartphones and Tablets

By getting smartphones and tablets to do the chunk of their work, smart little devices are able to sit snugly in watches and sunglasses, to lend a helping hand or a dash of excitement to your life! However, making such devices is challenging, considering the special technical, health, privacy, security and, well, fashion related concerns to be handled!

JANANI GOPALAKRISHNAN VIKRAM

Room-sized technology shrunk to desktop form factors, then to laptop and palmtop sizes, and now to such a level that you can wear it on your wrist, in your chain or as cool sunglasses. In fact, devices can even be weaved into your clothes or stitched on labels. Wearable

devices help in a variety of ways—to monitor the wearer's health, evaluate his exercise, diet or even sleep pattern, track location, deliver new gaming or travel experiences through augmented reality, or run simple applications.

"Most of the wearable technology spoken about today is either in the military domain or in the healthcare and related areas like fitness and sports. The features are mostly limited to basic con-

nnectivity using Bluetooth or Wi-Fi. Some of the other more advanced features like location information and augmented reality are in different stages of adoption. Augmented reality applications still need to see mainstream use cases. Location has a challenge of being battery-intensive, which affects adoption of these devices. However, these challenges are also likely to be overcome in a year or two and we will see wearable devices picking up in other areas as well," says Shivesh Vishwanathan, senior consultant (Mobility), Persistent Systems—a company that provides mobility solutions for wearable devices.

Vishwanathan cites a recent study, which estimates that 30 million mHealth wearable wireless devices were shipped in 2012—an increase of 37 per cent over shipments in 2011. Many of these devices are in the sports category from companies like Nike, Garmin and Adidas. Wearable device market is slated to rapidly increase and become a \$1-1.5 billion industry by 2014.

While some of these wearable tech accessories are street-smart and affordable, quite a few are associated with famous designers or fashion houses and cost many thousand dollars. The I'm Watch, for instance, is a smartphone companion that can handle calls, messages, emails, music and other applications, with an in-built ARM processor, customised Android operating system, a 4GB flash drive, 128MB RAM, battery, speakers, display and other features! It comes in various models priced from \$389 to \$19,990—and the designs vary from simple coloured



Vuzix Smart Glasses M100 with ear-mounting option (Courtesy: www.vuzix.com)